

Lean-4 verified geometric constraint yields charge-dependent mass spectrum

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A covariant geometric field model is analyzed under Lean 4 formal proof constraints. The framework satisfies positivity and conservation requirements necessary for producing dimensional predictions. The model introduces **no numerical parameters axiomatically**. Dimensional anchoring is achieved by setting the effective mass of a topological state with charge $Q = 1$ to $\mathbf{m}_{\text{eff}} \approx 1.0073 \times 10^{-27}$ kg. This normalization structurally fixes the dimensionless coupling ratio at $\mathbf{R} \approx 0.007304$, which is consistent with the fine-structure constant. Formal constraints impose a **UV Finiteness Lock**, limiting dynamically stable configurations to topological charges $\mathbf{Q} \leq 3$. The mass determination relies on the Topological Dominance constraint ($\mathbf{R} \equiv \mathbf{m}_{\text{ind}}/\mathbf{m}_{\text{topo}} \ll 1$). The resulting mass spectrum, verified by Lean 4, predicts two additional stable ground states at $\mathbf{Q} = 2$ ($\approx 0.5073 \times 10^{-27}$ kg) and $\mathbf{Q} = 3$ ($\approx 0.3403 \times 10^{-27}$ kg).

I. FORMAL CONSISTENCY AND CONSTRAINTS

A. Formal Consistency Check

The numerical predictions are supported by a formal structure verified through **Lean 4 formal certification**. This process establishes the logical consistency of explicit variational equations and validates key requirements, including **Positive Effective Mass** ($m_{\text{eff}} > 0$), conservation laws, and **Positive Topological Charge** ($Q > 0$). Stable particle ground states are structurally constrained by the ****UV Finiteness Lock**** ($Q \leq 3$).

B. Dimensional Anchoring and Coupling

The theory operates under the premise that **no numerical parameters are introduced axiomatically**. The physical scale is set by a single dimensional anchor, the $Q = 1$ effective mass, fixed at $\mathbf{m}_{\text{eff}} \approx 1.0073 \times 10^{-27}$ kg. This scale derivation yields the dimensionless coupling ratio $\mathbf{R} \approx 0.007304$.

II. DYNAMIC EQUATIONS AND MASS MECHANISM

A. Hydrodynamic and Dirac Mass Formulas

The field dynamics include conservation laws and a momentum balance equation, specified by the non-relativistic hydrodynamic limit:

$$\begin{aligned} \rho(\partial_t \mathbf{v} + (\mathbf{v} \cdot \nabla) \mathbf{v}) = & -\nabla P - \rho \nabla W(\Phi) \\ & - \kappa(\nabla^2 \Phi) \nabla \Phi + \lambda \nabla(\rho \Phi) \end{aligned} \quad (1)$$

The Dirac mass formula specifies mass as topologically determined:

$$m_{\Psi}(x) = \frac{T \ell_f}{c^2} \frac{1}{Q(x)} \quad (2)$$

The total measured mass (m_{eff}) is the sum of the Topological Mass (m_{topo}) and the Induced Inertial Mass (m_{ind}). For consistency and stability, the ****Topological Dominance constraint**** requires that the induced mass be significantly smaller than the topological mass ($\mathbf{R} \equiv \mathbf{m}_{\text{ind}}/\mathbf{m}_{\text{topo}} \ll 1$).

B. Gravitational Consistency

The calibrated effective mass is the physical input for the curvature term in the Total Action. Cross-sector consistency checks require the derived gravitational constant (G_{ind}) to match the Newtonian constant G . The induced gravitational constant scales as:

$$G_{\text{ind}} \sim \frac{c^4 \ell_f^2}{16\pi \mathcal{A} \kappa} \quad (3)$$

III. MASS SPECTRUM AND NOVEL PREDICTIONS

A. UV Finiteness Lock and Spectrum

Lean 4 verification confirms that the stability constraints necessitate the ****UV Finiteness Lock****, restricting dynamically stable filament configurations to $\mathbf{Q} \leq 3$. This constraint is essential for meeting the requirements of ****Positive Effective Mass**** and adherence to the ****Topological Dominance Condition****. The underlying mass spectrum is given by a quadratic string formula:

$$m_{\text{string}}^2 = k \cdot (N + \beta \cdot Q^2 - a) \quad (4)$$

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B. Predicted Stable Ground States

Based on the $Q \leq 3$ constraint and the dimensional anchor, the framework predicts two additional stable topological states ($N = 0$):

- **Q = 2:** $\approx 0.5073 \times 10^{-27}$ kg
- **Q = 3:** $\approx 0.3403 \times 10^{-27}$ kg

These particles are proposed as candidates for dark matter.

IV. INFORMATIONAL CLOSURE

The model attributes entanglement and collapse to worldline backbleed. Further work involves defining an entropy functional ($S_{\text{filament}} = -\int \rho_{\tau} \log \rho_{\tau} d\tau$) for the discrete twist ensemble (τ) to approach informational closure.

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The source implicitly suggests that the Lean 4 formal certification serves as a robust internal proof mechanism for the geometric field model.